Day4 Firewall Basic Theory

Task 1:- Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Follow these steps on your Windows computer.

Step 1: Open Windows Defender Firewall

1. Press `Windows + R`.
2. Type `wf.msc`.
3. Press Enter.
4. Select Outbound Rules from the left panel.

Step 2: Create a New Rule

1. Click New Rule on the right panel.
2. Select Program.
3. Click Next.

Step 3: Select Spotify Application

1. Select This program path.
2. Browse and locate `Spotify.exe`.

Common location:

```
%AppData%\Spotify\Spotify.exe
```

If Spotify is installed elsewhere, browse to its actual executable file.

Step 4: Block the Connection

1. Select Block the connection.
2. Click Next.
3. Select all three profiles:
    - Domain
    - Private
    - Public
4. Click Next.

Step 5: Name the Rule

Enter:

```
Block Spotify Outbound Traffic
```

Click Finish.

Expected result

The new outbound firewall rule should appear in the Outbound Rules list with its action set to Block.

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Task 2:- Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Step 1: Open Windows Defender Firewall

1. Press `Windows + R`.
2. Type `wf.msc` and press Enter.
3. Click Inbound Rules on the left panel.
4. Click New Rule.

Step 2: Select Rule Type

1. Select Program.
2. Click Next.
3. Select This program path.
4. Browse and select Google Chrome:

```
C:\Program Files\Google\Chrome\Application\chrome.exe
```

If Chrome is installed in another location, select its actual `chrome.exe` file.

Step 3: Configure the Protocol and Port

1. Select TCP.
2. Select Specific local ports.
3. Enter `443`.
4. Click Next.

Step 4: Allow the Connection

1. Select Allow the connection.
2. Click Next.
3. Select the required profiles:
    - Domain
    - Private
    - Public
4. Click Next.

Step 5: Name the Rule

Enter the following name:

```
Allow Chrome Inbound HTTPS 443
```

Click Finish.

Step 6: Verify the Rule

- Open Windows Defender Firewall with Advanced Security.
- Select Inbound Rules.
- Find `Allow Chrome Inbound HTTPS 443`.
- Check that the rule is Enabled and its Action is Allow.

Task 3:- Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender FirewallUse the 'Scope' tab when creating a new rule to specify the IP address

Step 1: Open Windows Defender Firewall

1. Press `Windows + R`.
2. Type `wf.msc`.
3. Press Enter.
4. Windows Defender Firewall with Advanced Security will open.

Step 2: Create a New Inbound Rule

1. Click Inbound Rules on the left panel.
2. Click New Rule on the right panel.
3. Select Custom.
4. Click Next.
5. Select All programs and click Next.

Step 3: Select Protocol

1. Under Protocol type, select Any.
2. Click Next.

Step 4: Configure the Scope Tab

1. Find the Scope section.
2. Under Which remote IP addresses does this rule apply to?, select These IP addresses.
3. Click Add.
4. Enter the IP address:

```
123.45.67.89
```

5. Click OK and then Next.

Step 5: Block the Connection

1. Select Block the connection.
2. Click Next.
3. Select all three profiles:
    - Domain
    - Private
    - Public
4. Click Next.

Step 6: Name the Rule

Enter the rule name:

```
Block Specific IP 123.45.67.89
```

Click Finish.

Step 7: Block Outgoing Traffic Too

To block outgoing traffic from your computer to the same IP:

1. Select Outbound Rules.
2. Click New Rule.
3. Repeat Steps 2–6 using the same remote IP address and select Block the connection.

Task 4:- Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Difference	Hardware Firewall	Software Firewall
1. Installation	Installed as a physical device between the network and the internet.	Installed directly on a computer or server.
2. Protection	Protects multiple devices across an entire network.	Protects an individual computer or device.
3. Performance	Uses dedicated hardware to handle network traffic.	Uses the computer's CPU and memory to filter traffic.

Use Cases

1. Hardware Firewall:

- Used in large offices, colleges, banks, and organizations.
- Protects the entire network from unauthorized access and cyberattacks.
- Example: Cisco ASA or Fortinet FortiGate.

2. Software Firewall:

- Used on personal computers and laptops.
- Controls incoming and outgoing network connections for individual applications.
- Example: Windows Defender Firewall.